A királyi őrség

2026-11-09

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(broom)
library(performance)
library(sjPlot)
```

```
##
## Attaching package: 'sjPlot'
##
## The following object is masked from 'package:ggplot2':
##
##     set_theme
```

```
library(knitr)
library(patchwork)
d = read_csv('https://raw.githubusercontent.com/petyaracz/AdatmanipulacioModellezes/main/dat/R2/kings_g
```

```
## Rows: 2000 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): province
## dbl (7): accepted, son, age, height, weight, score, grade
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

EDA

1. Nézzük meg az adatokat.

```
glimpse(d)
```

```
## Rows: 2,000
## Columns: 8
## $ accepted <dbl> 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 0, 0~
## $ son      <dbl> 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0~
```

```
## $ age      <dbl> 30, 16, 31, 47, 24, 18, 29, 20, 34, 73, 19, 24, 47, 63, 28, 2~
## $ height   <dbl> 176, 165, 193, 179, 176, 177, 183, 176, 180, 172, 169, 186, 1~
## $ weight   <dbl> 73, 61, 96, 84, 80, 90, 95, 74, 94, 77, 72, 91, 81, 58, 78, 6~
## $ score    <dbl> 51, 70, 43, 45, 64, 90, 65, 47, 55, 67, 82, 83, 45, 53, 56, 6~
## $ grade    <dbl> 3.65, 3.62, 2.72, 3.44, 3.36, 5.00, 2.73, 3.87, 3.34, 2.82, 4~
## $ province <chr> "east", "south", "east", "north", "west", "south", "east", "n~
```

```
summary(d)
```

```
##     accepted           son              age            height
##  Min.   :0.0000   Min.   :0.000   Min.   :16.00   Min.   :151.0
##  1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:19.00   1st Qu.:174.0
##  Median :0.0000   Median :0.000   Median :23.00   Median :178.0
##  Mean   :0.3255   Mean   :0.143   Mean   :28.28   Mean   :178.5
##  3rd Qu.:1.0000   3rd Qu.:0.000   3rd Qu.:31.00   3rd Qu.:183.0
##  Max.   :1.0000   Max.   :1.000   Max.   :80.00   Max.   :202.0
##      weight           score            grade           province
##  Min.   : 50.00   Min.   : 18.00   Min.   :2.070   Length   :2000
##  1st Qu.: 72.00   1st Qu.: 50.00   1st Qu.:3.380   N.unique :   4
##  Median : 80.00   Median : 59.00   Median :3.790   N.blank  :   0
##  Mean   : 79.74   Mean   : 59.82   Mean   :3.782   Min.nchar:   4
##  3rd Qu.: 88.00   3rd Qu.: 69.00   3rd Qu.:4.190   Max.nchar:   5
##  Max.   :129.00   Max.   :100.00   Max.   :5.000
```

Minden megfigyelés egy ember, aki jelentkezett a királyi őrségbe. Vajon felvették?

- accepted: felvették-e az embert az őrségbe
- son: az apja tagja a királyi őrségnek?
- height: testmagasság (cm)
- weight: súly (kg)
- score: hány pontot ért el a felvételi vizsgán
- grade: mi volt az iskolai átlaga
- province: melyik sarkából jött a királyságnak

2. Csináljuk meg szépen az adatokat.

A logikai változóknak adjunk címkéket. A folyamatos változókat skálázzuk és centralizáljuk a későbbi modellezéshez.

```
d = d |>
  mutate(
    accepted_factor = ifelse(accepted, 'accepted','rejected'),
    ancestry = ifelse(son, "guard's son", "commoner"),
    age_scaled = as.numeric(scale(age)),
    height_scaled = as.numeric(scale(height)),
    weight_scaled = as.numeric(scale(weight)),
    score_scaled = as.numeric(scale(score)),
    grade_scaled = as.numeric(scale(grade))
  )
```

3. Sétáljunk végig egyenként ezeken a változókon.

```
p1 = d |>
  ggplot(aes(y = ancestry,fill = accepted_factor)) +
  geom_bar(position = position_dodge()) +
  theme_bw() +
```

```
  scale_fill_grey() +
  ggtitle('ancestry')

p2 = d |>
  ggplot(aes(y = province,fill = accepted_factor)) +
  geom_bar(position = position_dodge()) +
  theme_bw() +
  scale_fill_grey() +
  ggtitle('province')

p3 = d |>
  ggplot(aes(accepted_factor, age)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('age')

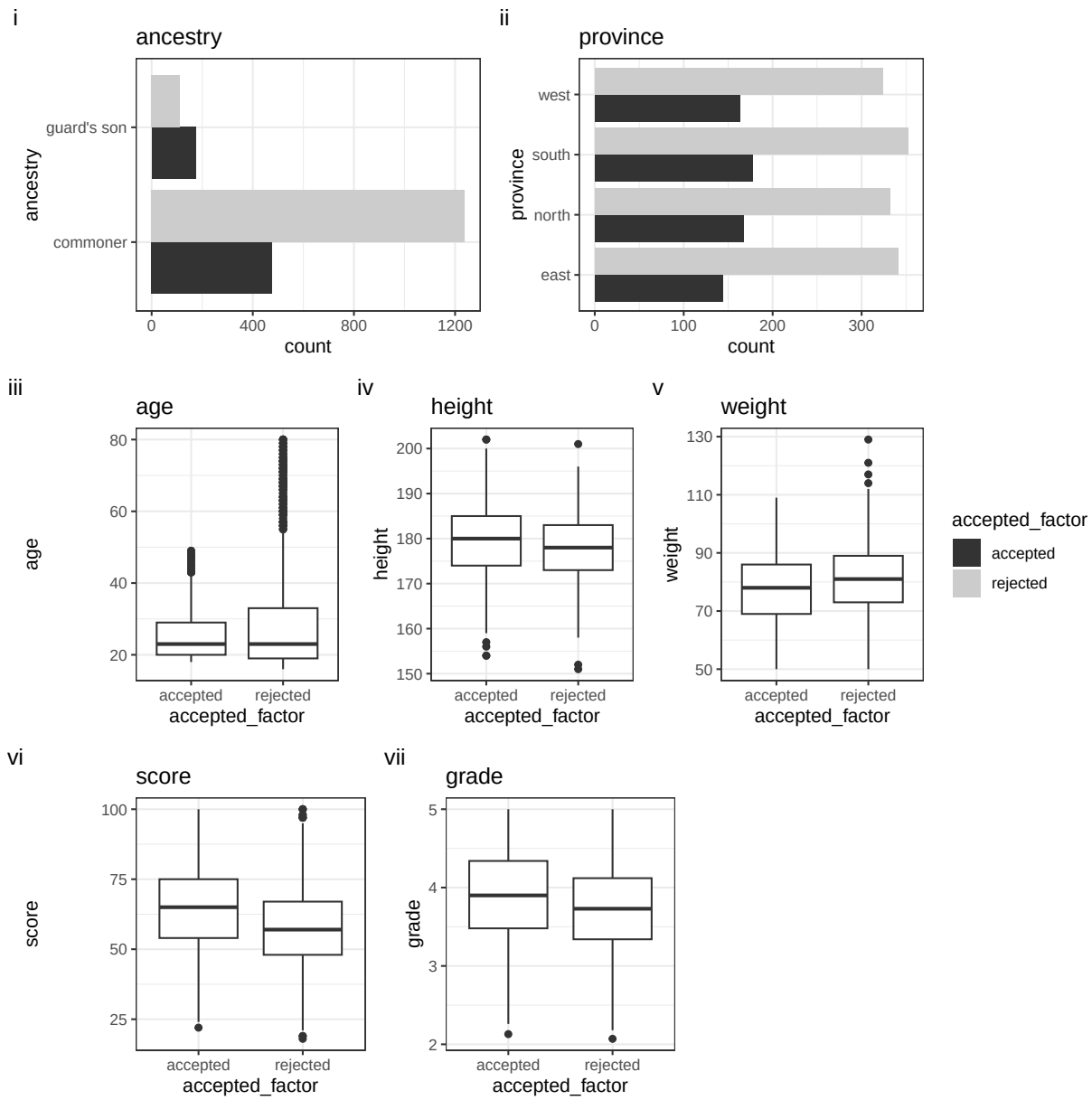
p4 = d |>
  ggplot(aes(accepted_factor, height)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('height')

p5 = d |>
  ggplot(aes(accepted_factor, weight)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('weight')

p6 = d |>
  ggplot(aes(accepted_factor, score)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('score')

p7 = d |>
  ggplot(aes(accepted_factor, grade)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('grade')

(p1 | p2 ) / ( p3 | p4 | p5) / (p6 | p7 | plot_spacer()) + plot_annotation(tag_levels = 'i') + plot_lay
```



Elég gyanúsak azok a bloxplotok.

```
p3 = d |>
  ggplot(aes(accepted_factor, age)) +
  geom_violin() +
  geom_boxplot(width = .1) +
  theme_bw()

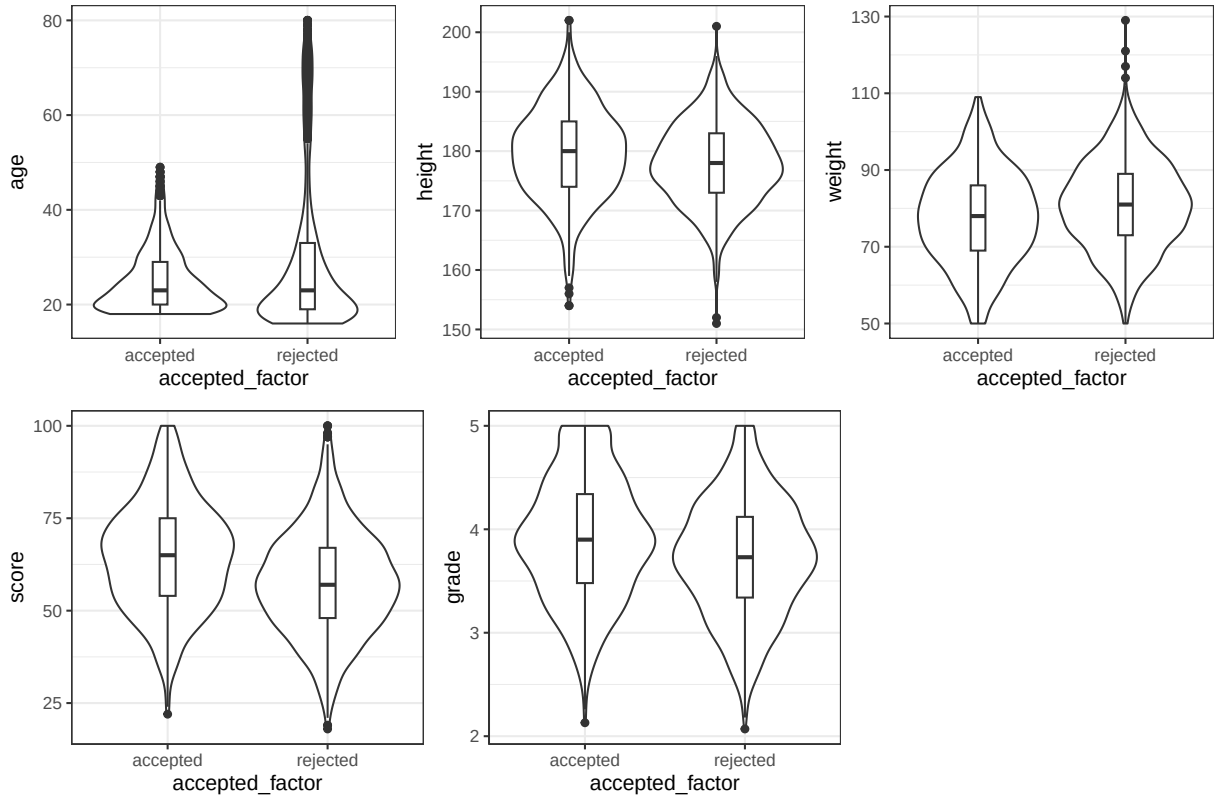
p4 = d |>
  ggplot(aes(accepted_factor, height)) +
  geom_violin() +
  geom_boxplot(width = .1) +
  theme_bw()
```

```
p5 = d |>
  ggplot(aes(accepted_factor, weight)) +
  geom_violin() +
  geom_boxplot(width = .1) +
  theme_bw()

p6 = d |>
  ggplot(aes(accepted_factor, score)) +
  geom_violin() +
  geom_boxplot(width = .1) +
  theme_bw()

p7 = d |>
  ggplot(aes(accepted_factor, grade)) +
  geom_violin() +
  geom_boxplot(width = .1) +
  theme_bw()

( p3 | p4 | p5) / (p6 | p7 | plot_spacer())
```

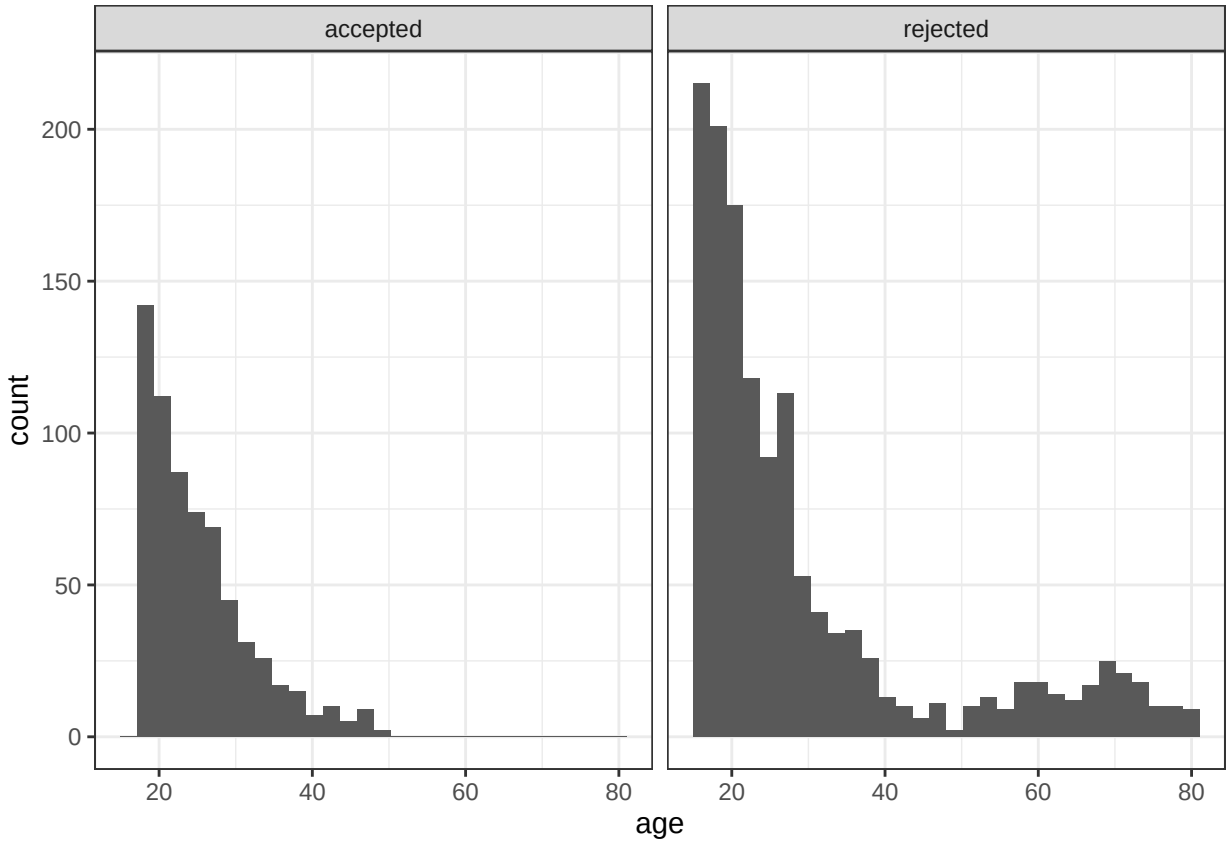


4. Az életkorral van valami kavarás.

```
d |>
  ggplot(aes(age)) +
  geom_histogram() +
  facet_wrap( ~ accepted_factor) +
```

```
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



18 éves kor alatt és 50 éves kor fölött nem vesznek fel senkit. Ezt külön kell kezelni, mert ezt egy lineáris modell nem kapja el.

```
d2 = d |>
  filter(
    age >= 18 & age <= 50
  )
```

Csináljuk meg újra a plotokat, most már csak azokra, akiknek egyáltalán van esélyük bekerülni.

```
p1 = d2 |>
  ggplot(aes(y = ancestry,fill = accepted_factor)) +
  geom_bar(position = position_dodge()) +
  theme_bw() +
  scale_fill_grey() +
  ggtitle('ancestry')

p2 = d2 |>
  ggplot(aes(y = province,fill = accepted_factor)) +
  geom_bar(position = position_dodge()) +
  theme_bw() +
  scale_fill_grey() +
  ggtitle('province')
```

```
p3 = d2 |>
  ggplot(aes(accepted_factor, age)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('age')

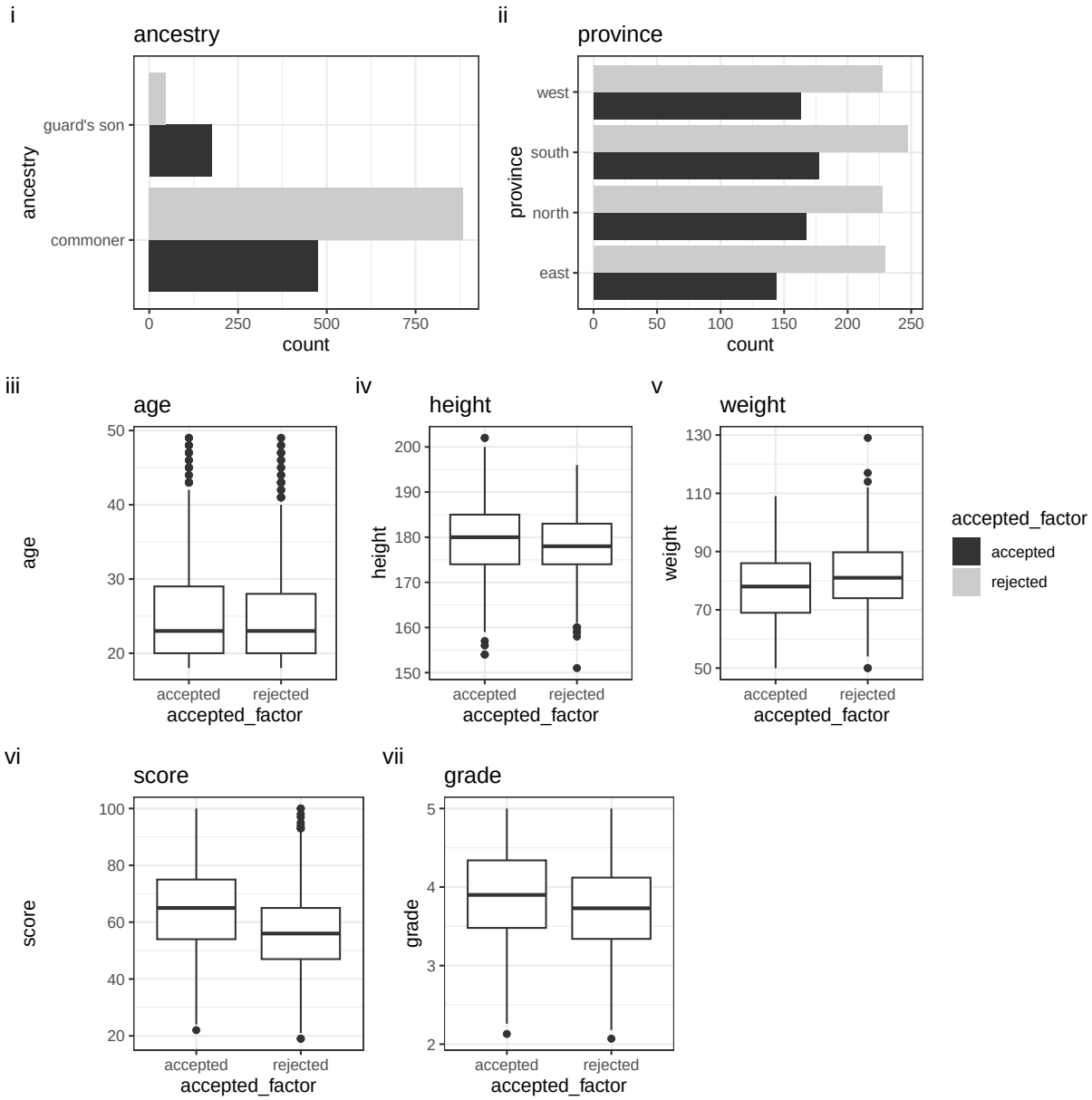
p4 = d2 |>
  ggplot(aes(accepted_factor, height)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('height')

p5 = d2 |>
  ggplot(aes(accepted_factor, weight)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('weight')

p6 = d2 |>
  ggplot(aes(accepted_factor, score)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('score')

p7 = d |>
  ggplot(aes(accepted_factor, grade)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle('grade')

(p1 | p2 ) / ( p3 | p4 | p5) / (p6 | p7 | plot_spacer()) + plot_annotation(tag_levels = 'i') + plot_lay
```



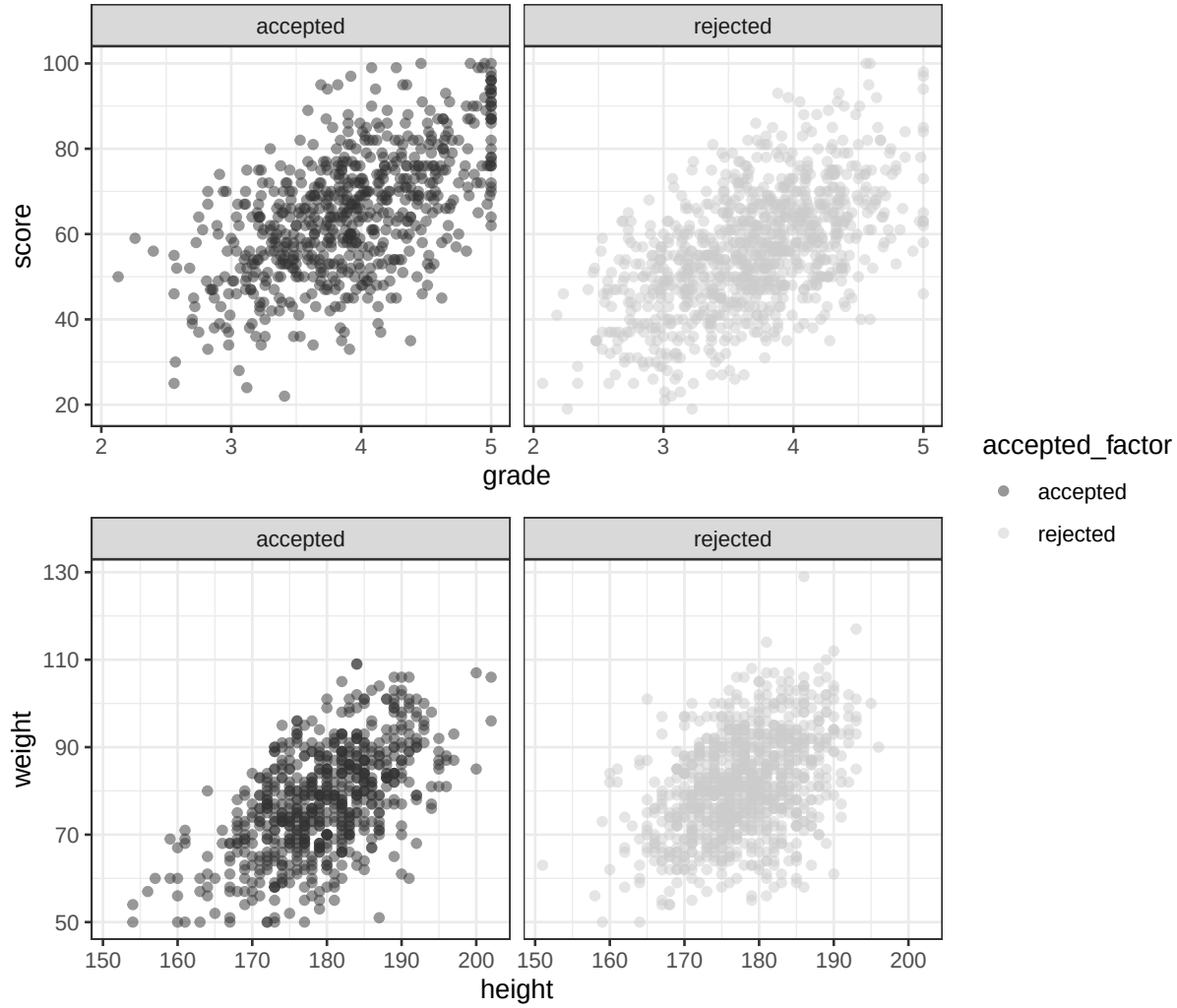
5. Vannak összefüggések a prediktorváltozók között?

```
p1 = d2 |>
  ggplot(aes(grade,score,colour = accepted_factor)) +
  geom_point(alpha = .5) +
  theme_bw() +
  facet_wrap( ~ accepted_factor) +
  scale_colour_grey()

p2 = d2 |>
  ggplot(aes(height,weight,colour = accepted_factor)) +
  geom_point(alpha = .5) +
  theme_bw() +
```

```
  facet_wrap( ~ accepted_factor) +
  scale_colour_grey()

p1 / p2 + plot_layout(guides = 'collect')
```



Hát itt elég sok megfigyelés van, nem megyünk ezzel sokra.

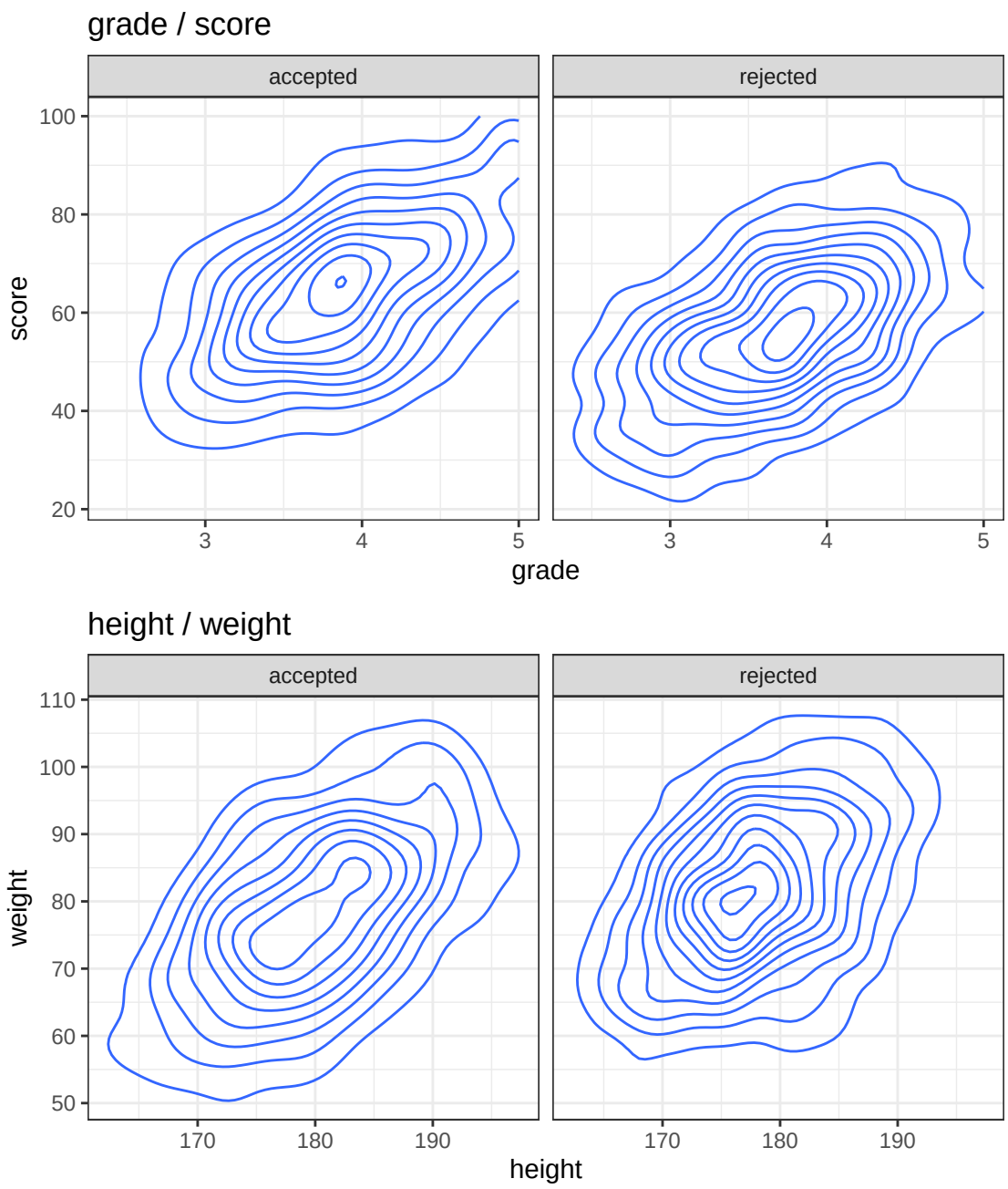
6. Vannak összefüggések a prediktorváltozók között? Máshogy nézzük.

```
p1 = d2 |>
  ggplot(aes(grade,score)) +
  geom_density_2d() +
  theme_bw() +
  facet_wrap( ~ accepted_factor) +
  ggtitle('grade / score')

p2 = d2 |>
  ggplot(aes(height,weight)) +
  geom_density_2d() +
```

```
  theme_bw() +
  facet_wrap( ~ accepted_factor) +
  ggtitle('height / weight')

p1 / p2
```



Modellezés

7. Mindent a levesbe.

```
glm1 = glm(accepted ~
             son +
             province +
             age_scaled +
             height_scaled +
             weight_scaled +
             score_scaled +
             grade_scaled,
           data = d2, # csak aki játszik
           family = binomial
             )

tidy(glm1, conf.int = T) |>
  kable(digits = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.76	0.13	-5.95	0.00	-1.01	-0.51
son	2.02	0.19	10.58	0.00	1.66	2.41
provincenorth	0.16	0.17	0.93	0.35	-0.17	0.49
provincesouth	0.06	0.17	0.37	0.71	-0.26	0.39
provincewest	0.06	0.17	0.37	0.71	-0.27	0.40
age_scaled	-0.02	0.13	-0.17	0.86	-0.27	0.22
height_scaled	0.46	0.07	6.66	0.00	0.33	0.60
weight_scaled	-0.63	0.07	-8.91	0.00	-0.77	-0.49
score_scaled	0.69	0.08	9.14	0.00	0.54	0.84
grade_scaled	0.06	0.07	0.90	0.37	-0.08	0.20

```
check_collinearity(glm1)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##           Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##            son 1.05 [1.02,  1.15]     1.03      0.95     [0.87, 0.98]
##       province 1.01 [1.00,  1.47]     1.01      0.99     [0.68, 1.00]
##     age_scaled 1.01 [1.00, 30.84]     1.00      0.99     [0.03, 1.00]
##  height_scaled 1.38 [1.31,  1.48]     1.18      0.72     [0.67, 0.76]
##  weight_scaled 1.39 [1.31,  1.49]     1.18      0.72     [0.67, 0.76]
##   score_scaled 1.49 [1.41,  1.60]     1.22      0.67     [0.62, 0.71]
##   grade_scaled 1.45 [1.36,  1.55]     1.20      0.69     [0.65, 0.73]
```

```
check_residuals(glm1)
```

```
## OK: Simulated residuals appear as uniformly distributed (p = 0.779).
```

```
check_overdispersion(glm1)
```

```
## # Overdispersion test (using simulated residuals)
##
##  dispersion ratio = 1.000
```

```
##           p-value = 0.976
```

```
## No overdispersion detected.
```

A modell okésan fest. A province, az age, és a grade nem csinál semmit.

8. Province, age, grade.

Az a korrekt, ha ezeket egyenként kidobáljuk, és megnézzük, romlik-e a modell.

```
glm2a = glm(accepted ~
              son +
              # province +
              age_scaled +
              height_scaled +
              weight_scaled +
              score_scaled +
              grade_scaled,
            data = d2, # csak aki játszik
            family = binomial
              )

glm2b = glm(accepted ~
              son +
              province +
              # age_scaled +
              height_scaled +
              weight_scaled +
              score_scaled +
              grade_scaled,
            data = d2, # csak aki játszik
            family = binomial
              )

glm2c = glm(accepted ~
              son +
              province +
              age_scaled +
              height_scaled +
              weight_scaled +
              score_scaled, # +
              # grade_scaled,
            data = d2, # csak aki játszik
            family = binomial
              )

compare_performance(glm1,glm2a, metrics = 'common')
```

```
## # Comparison of Model Performance Indices
##
## Name  | Model |  AIC (weights) |  BIC (weights) | Tjur's R2 |  RMSE
## ------------------------------------------------------------------
## glm1  |   glm | 1765.8 (0.072) | 1819.5 (<.001) |     0.231 | 0.432
## glm2a |   glm | 1760.7 (0.928) | 1798.2 (>.999) |     0.230 | 0.432
```

```
compare_performance(glm1,glm2b, metrics = 'common')
```

```
## # Comparison of Model Performance Indices
##
## Name  | Model |  AIC (weights) |  BIC (weights) | Tjur's R2 |  RMSE
## ------------------------------------------------------------------
## glm1  |   glm | 1765.8 (0.272) | 1819.5 (0.025) |     0.231 | 0.432
## glm2b |   glm | 1763.8 (0.728) | 1812.1 (0.975) |     0.231 | 0.432
```

```
compare_performance(glm1,glm2c, metrics = 'common')
```

```
## # Comparison of Model Performance Indices
##
## Name  | Model |  AIC (weights) |  BIC (weights) | Tjur's R2 |  RMSE
## ------------------------------------------------------------------
## glm1  |   glm | 1765.8 (0.355) | 1819.5 (0.036) |     0.231 | 0.432
## glm2c |   glm | 1764.6 (0.645) | 1812.9 (0.964) |     0.230 | 0.432
```

Nyugodt szívvel kidobálhatjuk ezt a három változót, nem csinálnak semmit.

9. Weight és Height

Van összefüggés a testsúly és a testmagasság hatása között? Illesszük újra az új, "minimalista modellt", és vessük össze a komplexebb változattal.

```
glm3a = glm(accepted ~
              son +
              height_scaled +
              weight_scaled +
              score_scaled,
            data = d2, # csak aki játszik
            family = binomial
              )

glm3b = glm(accepted ~
              son +
              height_scaled * weight_scaled +
              score_scaled,
            data = d2, # csak aki játszik
            family = binomial
              )

compare_performance(glm3a,glm3b, metrics = 'common')
```

```
## # Comparison of Model Performance Indices
##
## Name  | Model |  AIC (weights) |  BIC (weights) | Tjur's R2 |  RMSE
## ------------------------------------------------------------------
## glm3a |   glm | 1757.5 (<.001) | 1784.4 (<.001) |     0.229 | 0.432
## glm3b |   glm | 1728.6 (>.999) | 1760.8 (>.999) |     0.246 | 0.428
```

```
test_likelihoodratio(glm3a,glm3b)
```

```
## # Likelihood-Ratio-Test (LRT) for Model Comparison (ML-estimator)
##
## Name  | Model | df | df_diff | Criterion |  Chi2 |      p
## ---------------------------------------------------------
```

```
## glm3a |   glm |  5 |         |   1747.53 |       |
## glm3b |   glm |  6 |       1 |   1716.62 | 30.91 | < .001
```

```
test_bf(glm3a,glm3b)
```

```
## Bayes Factors for Model Comparison
##
##     Model                                                      BF
## [2] son + height_scaled * weight_scaled + score_scaled 1.29e+05
##
## * Against Denominator: [1] son + height_scaled + weight_scaled + score_scaled
## *   Bayes Factor Type: BIC approximation
```

```
check_collinearity(glm3b)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##                          Term  VIF   VIF 95% CI adj. VIF Tolerance
##                           son 1.04 [1.01, 1.15]     1.02      0.96
##                height_scaled 1.43 [1.35, 1.53]     1.20      0.70
##                weight_scaled 1.42 [1.34, 1.52]     1.19      0.70
##                 score_scaled 1.05 [1.02, 1.15]     1.03      0.95
##  height_scaled:weight_scaled 1.04 [1.01, 1.15]     1.02      0.96
##  Tolerance 95% CI
##      [0.87, 0.99]
##      [0.65, 0.74]
##      [0.66, 0.75]
##      [0.87, 0.98]
##      [0.87, 0.99]
```

Oksa. Megvan a legjobb modell. Ábrázoljuk.

10. Oksa. Megvan a legjobb modell. Ábrázoljuk.

```
p1 = plot_model(glm3b, 'pred', terms = c('height_scaled','weight_scaled')) +
  theme_bw() +
  scale_fill_grey() +
  scale_colour_grey()
```

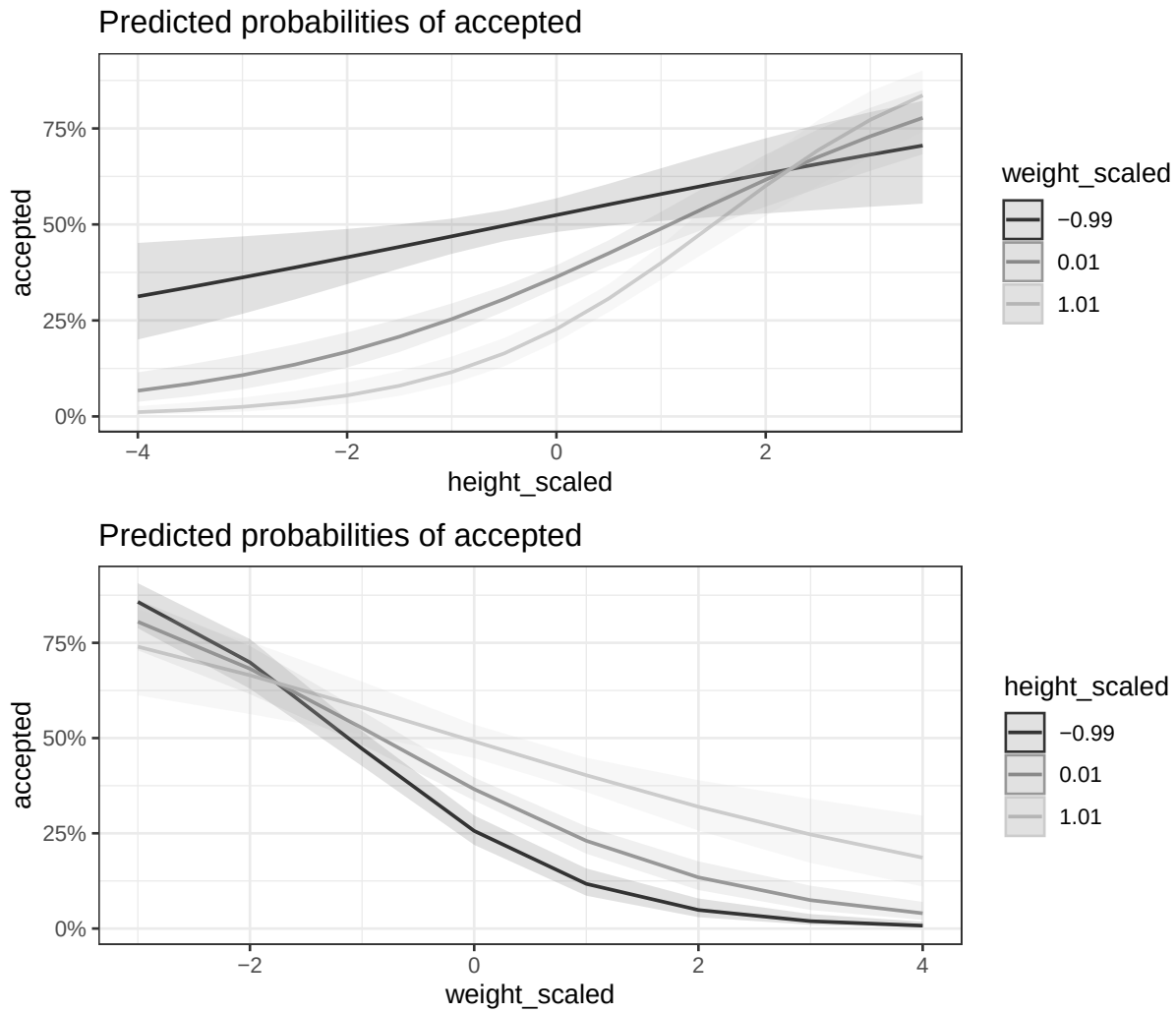
```
## Data were 'prettified'. Consider using `terms="height_scaled [all]"` to
##   get smooth plots.
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

```
p2 = plot_model(glm3b, 'pred', terms = c('weight_scaled','height_scaled')) +
  theme_bw() +
  scale_fill_grey() +
  scale_colour_grey()
```

```
## Data were 'prettified'. Consider using `terms="weight_scaled [all]"` to
##   get smooth plots.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

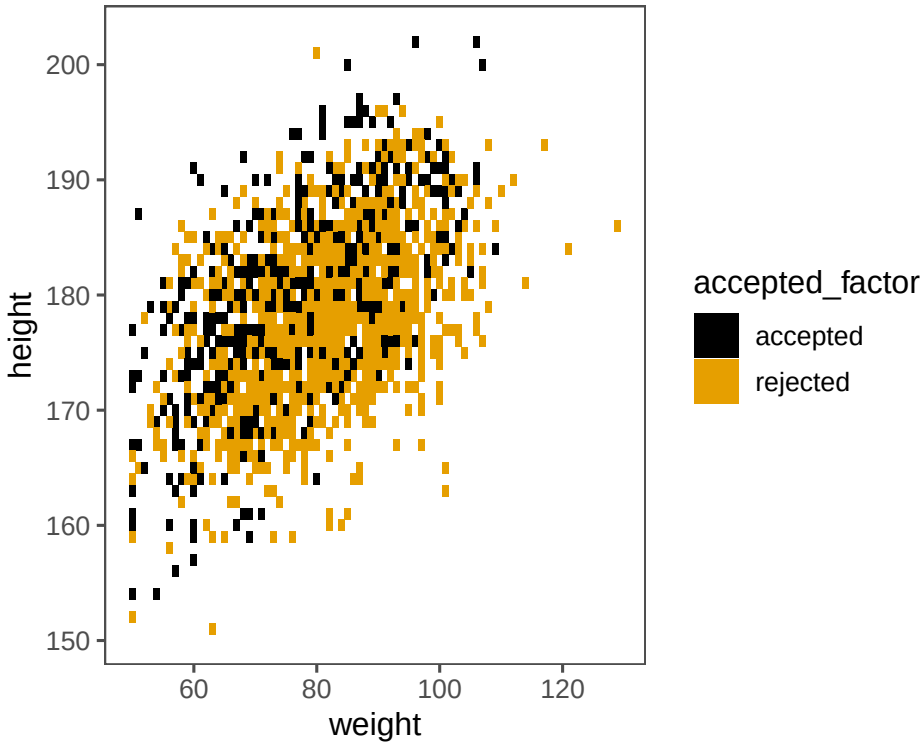
```
p1 / p2
```



11. Ezt az interakciót még kicsit emésztgetni kell.

```
d |>
  ggplot(aes(weight,height,fill = accepted_factor)) +
  geom_tile() +
  ggthemes::theme_few() +
  ggthemes::scale_fill_colourblind()
```

```
## Warning: `scale_fill_colorblind()` was deprecated in ggthemes 5.2.0.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```


height
200
190
180
170
160
150
60
80
100
120
weight
accepted_factor
accepted
rejected